

Hub Motor

Technical Data Sheet



Overview

The Mecwin 12-Inch Hub Motor is a robust and efficient BLDC motor designed for modern electric mobility solutions. With its integrated wheel design, it provides smooth performance, high torque, and low maintenance, making it perfect for electric scooters, e-bikes, and lightweight EVs. Built using advanced technology and durable materials, it ensures reliable operation, energy efficiency, and long service life—demonstrating Mecwin’s commitment to innovation and sustainable mobility.



Motor Specification :

Motor Specification			
Parameters	Specifications	Parameters	Specifications
Operating Voltage	60 V	Rated Torque	44 Nm
Rated Power	3.0 kW	Peak Torque	131 Nm
Peak Power	3.6 kW	Max Speed	714 Rpm
Max. DC Current	70 A		

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Parameters	Specifications	Parameters	Specifications
Outer Diameter	332 mm	Break Type	130 mm Drum
Shaft Length	275 mm	Swing Arm Distance	192.5 mm
IP Level	IP67	Application	2 wheeler

Performance Details :

Parameter	DC Voltage	DC Current	Input Power	Torque	Speed	Output Power	System Efficiency
	V	A	W	N-m	RPM	W	%
No Load	60	4.0262	241.47	0	714.7	2.9576	1.2
Max Torque	60	67.13	4025.6	131.3	90.6	1245.1	30.9
Max Input Power	60	70.808	4246	70.4	473.1	3487.6	82.1
Max Output Power	60	70.102	4203.8	62.2	552	3594.4	85.5
Max Efficiency	60	26.385	1582.4	20	700	1463.9	92.5

For More Information, Contact Mecwin :

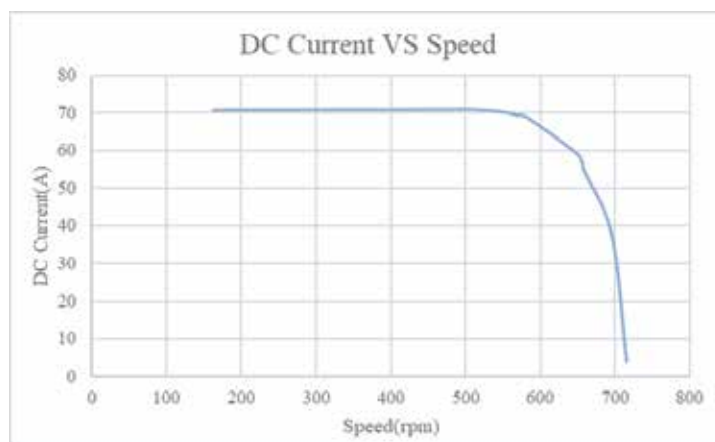
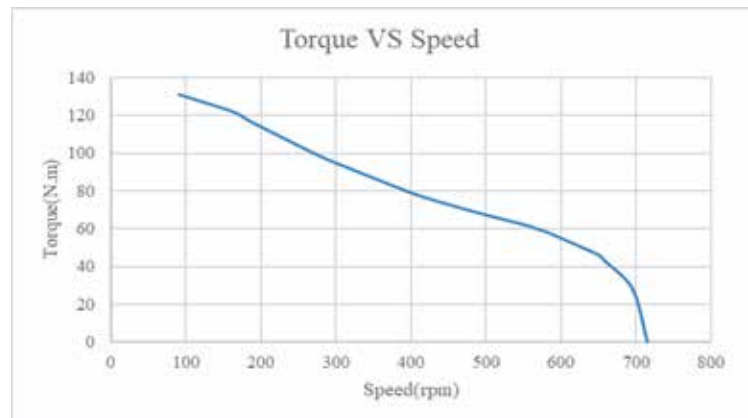
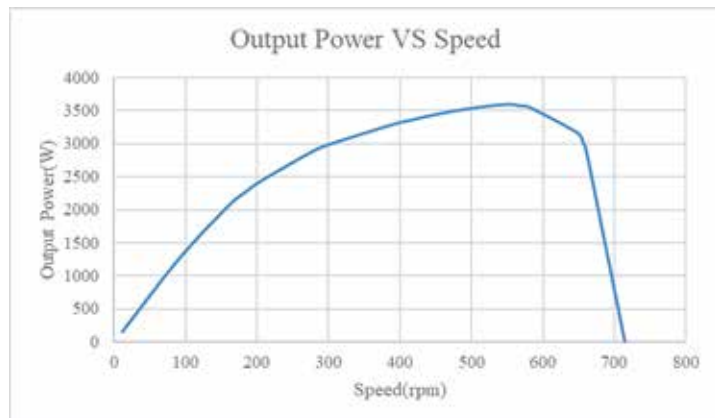
Tel : +91 80500 84932, +91 81239 67692

Email : sales@mecwinindia.com

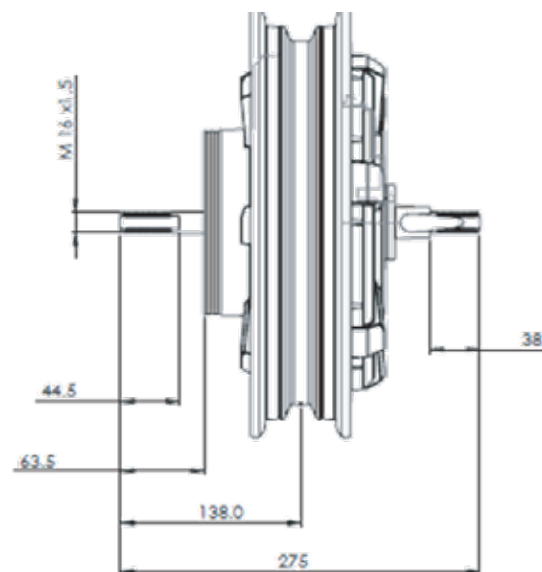
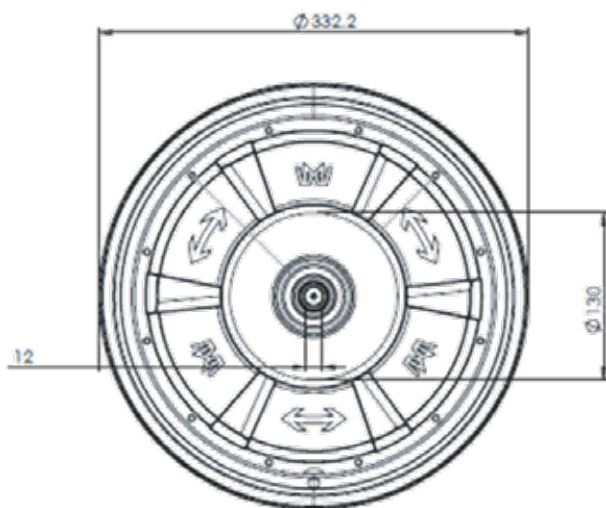
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Performance Curves :



Mechanical 2D Diagram :



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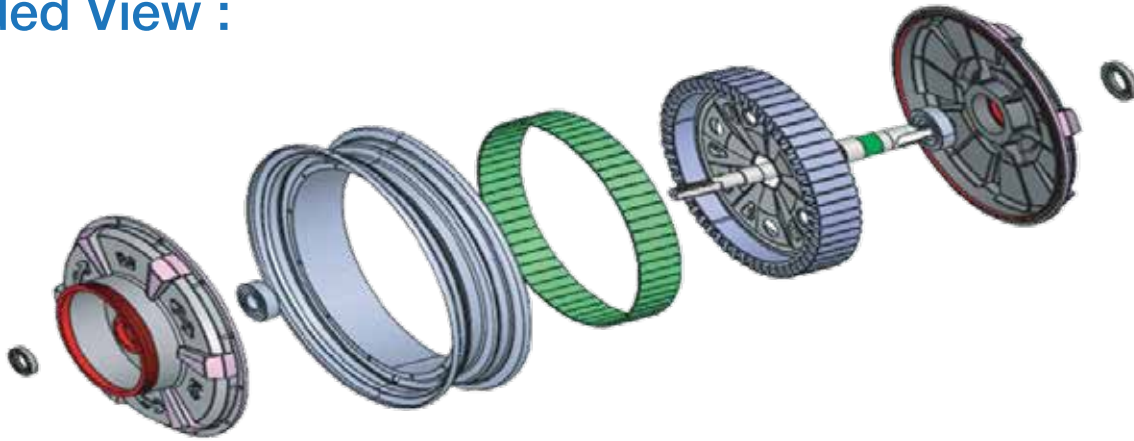
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Exploded View :



Technical Data :

DC Voltage	DC Current	Input Power	Torque	Speed	Output Power	System Efficiency
(V)	(A)	(W)	(N-m)	(RPM)	(W)	(%)
59.975	4.0262	241.47	0	714.9	2.9576	1.2
59.979	4.4439	266.54	0	714.7	2.6607	1
59.969	3.9527	237.04	0	713.5	1.2131	0.5
59.962	4.2583	255.33	0.1	713	9.8619	3.9
59.972	4.7207	283.11	0.4	712.9	29.949	10.6
59.965	4.8454	290.56	0.9	712.1	65.804	22.6
59.966	5.5947	335.49	1.5	709.2	113.4	33.8
59.985	18.519	1110.9	13	699.5	953.1	85.8
59.98	22.016	1320.5	16.2	697.9	1184.3	89.7
59.974	26.249	1574.3	19.9	696.6	1456.8	92.5
59.974	26.385	1582.4	20	696	1463.9	92.5
59.972	31.271	1875.4	23.6	695.4	1717.9	91.6
59.971	37.434	2244.9	28	695.4	2036.3	90.7
59.968	53.689	3219.6	42.4	660.2	2932.9	91.1
59.967	55.268	3314.3	43.6	656.9	3000	90.6
59.966	58.99	3537.4	46.5	649.1	3158.1	89.3
59.967	68.782	4124.6	58.3	581.5	3550.1	86.1
59.967	69.357	4159.1	60	568.6	3569.4	85.8
59.967	70.102	4203.8	62.2	552	3594.4	85.5
59.966	70.767	4243.6	66.1	513.4	3552.5	83.7
59.965	70.804	4245.8	70	476.8	3493.6	82.2
59.965	70.808	4246	70.4	473.1	3487.6	82.1
59.965	70.765	4243.4	74.4	438	3411.7	80.4
59.964	70.741	4242	77.6	411.8	3345.4	78.9
59.965	70.744	4242.2	80	394.8	3303.4	77.9
59.965	70.734	4241.6	95.3	298	2974.1	70.1
59.964	70.675	4237.9	100	271.6	2842.7	67.1
59.964	70.626	4235	113.6	202.9	2412.3	57
59.965	70.646	4236.3	117.7	182.7	2251.5	53.1
59.965	70.591	4233	122.2	162.6	2080.5	49.1
59.968	67.13	4025.6	131.3	90.6	1245.1	30.9
59.974	41.638	2497.2	120.3	11.8	148.23	5.9

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